

Proposal for

FINAL YEAR PROJECT IN AI
Umm Al Qura University



Wali

Project Title: Wali: An AI-Based Proactive System for Preventing Domestic Child Accidents

Team Members:

Student Name	Student ID	Email	Role
Rema Saed Althaqfi			Leader
Reema Salem Alharbi			Member
Enas Abdullah Saud			Member
Noran Abdullah Aljodi			Member

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Background

This project lies in the field of Computer Science, specifically within Computer Vision and Artificial Intelligence, and focuses on child safety in home environments. Children are naturally curious and may unknowingly place themselves in risky situations due to changes in their surroundings, while caregivers cannot always monitor them continuously. Previous work has used vision-based systems to monitor children or analyze their environment, but most approaches treat these aspects separately rather than considering their interaction. Recent AI systems can analyze video in real time and detect environmental changes, yet there is still a need for integrated solutions that monitor both the child and their surroundings simultaneously, providing timely support to caregivers before accidents occur.

Project Scope

This project involves the design and development of an AI-based smart system for child protection. The system will provide automated monitoring and generate real-time alerts when potential risks or unsafe situations are detected. The proposed solution relies on artificial intelligence technologies to recognize environmental changes and detect abnormal or unsafe conditions. In addition to the detection process, a user-friendly dashboard interface will be integrated to display system decisions and alerts, allowing caregivers to clearly monitor system outputs.

Project Description

Despite recent advances in computer vision-based monitoring systems, the majority of existing child safety solutions focus on either monitoring the child's activity or detecting environmental hazards as separate tasks. Many current systems rely on passive monitoring or post-incident alerts, which limits their ability to prevent accidents before they occur. There is still a research and development gap in integrated, real-time systems that simultaneously analyze the child's position, behavior, and proximity to potential hazards in the environment. Addressing this gap is critical for enabling proactive safety interventions rather than reactive ones.

The primary goal of this project is to develop an AI-based child safety monitoring system that proactively detects unsafe situations by analyzing both the child and the surrounding environmental risks. The proposed system employs computer vision and, when necessary, deep learning techniques to identify hazardous objects—such as sharp tools or fire—and to estimate their spatial relationship to the child in real time. Based on this analysis, the system generates timely alerts and notifications through a user-friendly interface, enabling caregivers to take immediate action. The system will include a simple dashboard for visually displaying detected risks and generated alerts for evaluation purposes. The project aims to design and evaluate this integrated approach as a functional system developed for research and testing, demonstrating the feasibility and effectiveness of proactive child safety monitoring in domestic settings.

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Aims and Objectives

The aim of this project is to design and develop an AI-based smart child protection system that enhances child safety by continuously monitoring the child and the surrounding environment and providing timely alerts to caregivers when potential risks are detected.

These aims will be achieved through the following objectives:

- The development of an intelligent monitoring system using computer vision to observe the child and the surrounding environment in real time.
- The application of machine learning techniques for the detection of unsafe situations, abnormal behaviors, and potential hazards.
- The integration of an alert and notification mechanism to immediately inform caregivers when intervention is required.
- The provision of a simple and user-friendly interface that enables caregivers to easily monitor system outputs and alerts.
- The evaluation of the overall performance and effectiveness of the proposed system under different conditions.
- The presentation of the proposed system concept to demonstrate the feasibility and practicality of the suggested solution.

Expected Outcomes

- **AI Detection Model:** Development of a high-accuracy AI-based model for real-time detection of children and hazardous objects.
- **Proactive Risk Analysis:** A software module to estimate distances between children and hazards to assess and predict potential risks.
- **System Integration:** Seamless integration with smart camera hardware for continuous real-time video processing.
- **Instant Alert System:** An automated notification interface to immediately alert caregivers through supported platforms.
- **Privacy-First Architecture:** An edge-based processing framework to ensure privacy and local video analysis.
- **Performance Evaluation:** A documented evaluation of system accuracy, latency, and false alarm rates.
- **User Interface:** A simple dashboard or application to display real-time alerts and system outputs for caregivers.

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Method/Approach

The AI-Based Proactive System for Preventing Domestic Child Accidents is developed following the CRISP-DM methodology (Schröder et al., 2021), a structured framework for data-driven and machine learning projects. This methodology provides a systematic and iterative process to develop, evaluate, and improve the system.

1. Business Understanding

The system aims to detect hazardous situations involving children in home environments and provide timely alerts to caregivers, reducing the risk of accidents.

2. Data Understanding

A custom dataset of images and videos representing children in typical home environments is collected. The dataset includes variations in lighting, room types, object categories, and camera angles to ensure model robustness.

3. Data Preparation

The dataset is labeled to identify children and potential hazards such as sharp tools, and heat sources. Preprocessing includes resizing, normalization, annotation formatting, and data augmentation. The data is divided into training, validation, and testing sets.

4. Modeling

A YOLO-based computer vision model is applied for real-time detection of children and hazardous objects ([Liang, 2024](#)). Euclidean distance calculations estimate the proximity of children to hazards ([Rahmatillah, 2023](#)). Deep learning techniques are further utilized to improve detection in complex scenarios such as occlusions or challenging lighting ([Alzubaidi et al., 2021](#)).

5. Evaluation

Model performance is assessed using Accuracy and mean Average Precision, with attention to minimizing false positives and false negatives to ensure reliable real-time operation.

6. Deployment

The trained model is integrated into a monitoring system that continuously analyzes video frames, evaluates the child's proximity to hazards, and provides immediate alerts through a user-friendly interface to support proactive caregiver intervention.

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Relevant references

1. Schröer, C., Kruse, F., & Gómez, J. (2021). A systematic literature review on applying CRISP-DM process model. *Procedia Computer Science* [CRISP-DM Process Model -](#)
2. Liang, J. (2024). *A review of the development of YOLO object detection algorithm*. *Applied Computer Engineering*, 71(1), 39–46. [YOLO](#)
3. Rahmatillah, S. (2023). *Webcam-Based Physical Distancing Prototype to Calculate Real-Time Distance by Applying Euclidean Distance*. [Euclidean Distance](#)
4. Alzubaidi, L., Zhang, J., Humaidi, A. J., Al-Din, Z., Al-Kabi, M., Santamaría, J., ... & Bouridane, A. (2021). *Review of deep learning: concepts, CNN architectures, challenges, applications, future directions*. *Journal of Big Data*, 8, Article 53. [Review of deep learning](#)